MATH118 Homework Assignment 1

DUE

October 13, 2025, 23:55

Description

- This is an individual assignment. Please do not collaborate. Don't write anything you cannot explain.
- If you think that this document does not clearly describe the assignment, ask questions before its too late.

Instructions:

- Show all steps clearly.
- Include brief written explanations for your reasoning.
- You may discuss with classmates, but your submitted work must be your own.
- Illegible or identical-looking submissions will not be graded.

Questions

Question 1 – Probability of Combined Events

Consider two events A and B with the following probabilities:

$$P(A) = 0.4, \quad P(B) = 0.5, \quad P(A \cup B) = 0.7.$$

Answer the following:

1. Compute $P(A \cap B)$.
2. Determine whether A and B are **mutually exclusive**. Explain why or why not.
3. Determine whether A and B are **independent**. Show your reasoning.
4. Find $P(A' \cap B)$ and $P(A' \cup B')$.

Show all steps clearly and provide brief justifications for your results.

Question 2 – Conditional Probability and Bayes' Theorem

In a quality control process, 4% of all manufactured items are defective.
A testing machine is used to identify defective items.
If an item is defective, the test correctly identifies it as defective with probability 0.92.
If the item is not defective, the test incorrectly marks it as defective with probability 0.03.

1. What is the probability that a randomly chosen item is marked as defective by the test?
   (Use the **Law of Total Probability**.)
2. If the test marks an item as defective, what is the probability that it is actually defective?
   (Use **Bayes' Theorem**.)
3. Interpret your result: is the testing process reliable?

Show all steps clearly and simplify your final results.

Question 3 – Counting, Conditional Probability, and Independence

An inspector randomly selects **3 parts without replacement** from a batch that contains 5 defective parts and 15 good parts (20 parts total). All selections are equally likely.

1. What is the probability that **exactly 2** of the selected parts are defective?
   (Show the counting argument using combinations.)

2. Given that **at least one** selected part is defective, what is the probability that **all three** selected parts are defective?
   (Compute the conditional probability $P(\text{all 3 defective} \mid \text{at least 1 defective})$.)
3. Let event (A) = "the first selected part is defective" and event (B) = "the second selected part is defective."
   a. Compute (P(A)) and (P(B)).
   b. Compute $P(A \cap B)$.
   c. Are (A) and (B) independent? Justify your answer with computations.

Show all steps clearly and simplify your final answers.

Submission Instructions

- This is a **handwritten assignment**.
- Use **pen or pencil** on clean, lined or plain A4 paper.
- Write your **name**, **student ID**, and **section** clearly on the **first page**.
- Show all steps, calculations, and justifications neatly.
- Maximum 1 page for each question; make sure they are numbered and ordered properly.
- After completing the work, **scan or photograph** your pages clearly and compile them into a **single PDF file**.
- The file name should follow this format: `first_name_last_name_HW1.pdf`
- No other format will be accepted.

Late Submission

- Not accepted.

Grading (Tentative)

- `Max Grade : 100`
- If you don't attend the related quiz, your homework submission will be ignored. It will be considered as 'not submitted'.